ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

- **English Name:** Luong Quynh Anh
- **Chinese Name:** 良琼英
- **Email:** 115202165@nccu.edu.tw
- **Student ID:** 115202165
- **University:** National Chengchi University

Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. There are 594 documents in this specific category.
2. Isolating specific categories before applying algorithms allows the algorithms to focus on documents from a specific domain. This reduces the noise from other unrelated domains and can help discover meaningful patterns. For instance, isolating the category "sci.med" before frequent pattern mining can make the algorithm reveal frequent medical terms rather than patterns from unrelated categories.

AI used for this answer: Yes

If yes, how: explaining frequent pattern mining, brainstorming

External references (links): https://www.geeksforgeeks.org/dsa/frequent-pattern-mining-in-data-mining/

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Because in that case, we will be missing out on nearly half of the population, which is indeed a huge number and is not beneficial to us in this era of scarce data. A small population often shows that our data is not representative enough to be used in further research. We can try categorization/classification methods to estimate the missing values, i.e. assign a category name for the uncategorized documents.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. The category proportions in the sample do not 100% match the original but are very comparable, in detail:

- *Category: alt.atheism, Original: 480, Sampled: 214, Proportion in original: 21.27%, Proportion in sample: 21.40%*
- *Category: soc.religion.christian, Original: 599, Sampled: 248, Proportion in original: 26.54%, Proportion in sample: 24.80%*
- *Category: comp.graphics, Original: 584, Sampled: 278, Proportion in original: 25.88%, Proportion in sample: 27.80%*
- *Category: sci.med, Original: 594, Sampled: 260, Proportion in original: 26.32%, Proportion in sample: 26.00%*

2. The minority class risks being omitted or under-represented in the sample. So, when we run models with this sample, the models might be biased towards the majority class and provide limited results for the minority class.

AI used for this answer: No

If yes, how:

External references (links): https://medium.com/dataman-in-ai/handbook-of-anomaly-detection-15-sampling-techniques-for-extremely-imbalanced-data-426145208fdb

---

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The number of features increased from 35482 to 35788.
2. The overall sparsity of the matrix will decrease. As stopwords appear in almost all documents, the number of 0's in the matrix will decrease and so will the overall sparsity.

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5059
0.05	1491

1. The total size of vocabulary decreases as we increase the threshold.
2. Very rare words, specific code/ID and abbreviations are the kinds of words which have variance close to zero and are eliminated first. This is because those words mostly appear in only 1 document but not the others, causing the distribution to be like [0 0 0 1 0 0 0], for example.
3. Because words with near-zero variance, as I said in *2.*, are usually rare words which cannot represent a category. These words mostly add noise to the model so reducing dimensionality by eliminating them does not have a big impact on model's accuracy.

AI used for this answer: No

If yes, how:

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

4. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
5. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
6. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

4. Most of the top-correlated words are domain-specific terms for graphics, such as: graphics, files, keywords, vga, etc.

5. Pearson correlation measures the strength of a linear relationship between the original numerical values of two variables; whereas Spearman correlation measures the strength of a monotonic relationship using the ranks of the values.

6. This would happen when the word correlates positively with the binary indicator, but not in a constant way, i.e. the points on the scatter plot (of these two variables) don't exactly form a straight line going up, but they still concentrate around that line.

AI used for this answer: No

If yes, how:

External references (links): https://support.minitab.com/en-us/minitab/help-and-how-to/statistics/basic-statistics/supporting-topics/correlation-and-covariance/a-comparison-of-the-pearson-and-spearman-correlation-methods/

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Maximal Frequent patterns were generated successfully using MaxFp-Growth algorithm

Total No of patterns: 710

Runtime: 1.1601769924163818

Using minimumSupportCount: 9 and extracting patterns for category: alt_atheism

Maximal Frequent patterns were generated successfully using MaxFp-Growth algorithm

Total No of patterns: 9547

Runtime: 2.644808292388916

Using minimumSupportCount: 3 and extracting patterns for category: alt_atheism

Decreasing minSup from 9 to 3 increases the execution runtime and the total number of discovered patterns. This is because when minSup is lower, more patterns meet the frequency condition as they only need to appear a small number of times.

Answer (2) — Top-K k vs MaxFPGrowth minSup: The parameter *k* in Top-K FAE gives us the **top *k*** most frequently appearing patterns, while minSup gives us the patterns that appear at least ***minSup*** times (i.e. appear a number of times equal to the minSup).

Answer (3) — Maximal vs standard frequent patterns: Maximal frequent patterns are structurally more concise patterns because they only keep frequent patterns that have no frequent supersets. Smaller frequent patterns that are subsets of larger frequent patterns are excluded, reducing redundancy and the total number of patterns.

AI used for this answer: Yes

If yes, how: explaining maximal frequent patterns and helping shaping the answers from my idea.

External references (links): https://gds.techfak.uni-bielefeld.de/_media/teaching/2020summer/bda/frequentitemsets.pdf

---

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. Because the raw **X_counts** matrix contains, in each row a document, in each column a word, and in each cellule how many times the word appears. It may contain tens of thousands of features depending on the number of words, whereas a 2D scatter plot can display only two dimensions. Therefore, the original matrix cannot be plotted directly in its full dimensionality.
2. Dimensionality reduction algorithms, such as PCA, UMAP or t-SNE, as referenced in the *visualization_helpers.py* file

AI used for this answer: Yes

If yes, how: brainstorming and explaining the idea

External references (links): https://towardsdatascience.com/the-achilles-heel-of-scatter-plots-f5e0a40b5548/

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. The plot produced by t-SNE/UMAP shows more distinct cluster islands for the 4 categories. We can see all four categories clearly in these two plots. On the other hand, in the PCA plot, the dots heavily overlap each other in the middle (around the 0 axis), so we cannot see all categories clearly.
2. ACP is a linear dimension reduction method that aims to maximize the variance captured by the selected components, rather than directly separating documents by category. As a result, the first two or three principal components may not reflect the complex, nonlinear relationships necessary to distinguish between high-dimensional text categories. In contrast, the t-SNE and UMAP methods can capture these nonlinear structures, which can provide clearer visual clusterings.
3. Because the t-SNE method primarily preserves local similarities rather than global distances. The perceived separation between clusters in the 2D plot may not reflect their actual dissimilarity in the original high-dimensional space.

AI used for this answer: Yes

If yes, how: brainstorming, word utilization check

External references (links): https://www.researchgate.net/publication/396225637_Comparative_Study_of_PCA_t-SNE_and_UMAP_for_CNN_Feature_Representation_of_Image_Classification

https://sklearn.org/1.6/auto_examples/manifold/plot_t_sne_perplexity.html

---

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4 binary columns.
2. Using the raw integer values (0, 1, 2, 3) makes the algorithm treat the categories as if they had a natural order and meaningful numerical distances, while they are just category labels and have no signification.
3. One-hot encoding represents each category using a separate binary column, rather than a single integer. Each category is assigned to a unique vector, without implying any meaningful numerical order among the categories. This allows algorithms to distinguish between categories without treating their labels as ordered numerical values.

AI used for this answer: Yes

If yes, how: brainstorming, giving example on one-hot binarization to allow deeper understanding.

External references (links): https://machinelearningmastery.com/one-hot-encoding-understanding-the-hot-in-data/
